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BUREAU OF SHIPS MANUAL

CHAPTER 24

SHIP CONTROL EQUIPMENT

NAVSHIPS 250-000-24



U. S. NAVY DEPARTMENT

WASHINGTON 25, D. C.

BUREAU OF SHIPS MANUAL

CHAPTER 24 SHIP CONTROL EQUIPMENT

**NAVY DEPARTMENT,
Bureau of Ships,
1 August 1946.**

This chapter is a revision of Bureau of Ships Manual, Chapter 24, "Ship Control Equipment," dated 4 October 1944.

This revised chapter becomes effective upon receipt and shall be inserted in its proper place in the Manual binder.

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Chapter 24

SHIP CONTROL EQUIPMENT

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Section I. Navigational Instruments

Part 1—Gyrocompass Equipment

24-1.

Gyrocompasses and their power-generating units, both main and auxiliary, supply panels, control panels, navigation and ship-control repeater panels, navigation and ship-control repeater compasses, dead-reckoning equipments, self-synchronous alidades, plotting facilities, (which include horizontal plotting tables Mk. 1; summary plotting boards 24" (synchro) Mk. 2; summary plotting boards 24" (step-by-step) Mk. 3; vertical plotting boards 60" Mk. 4; edge-lighted status boards 36" Mk. 5; vertical plotting boards 36" Mk. 6; and visual fighter director plotting board 24" (synchro) Mk. 7) and all proprietary repair parts for these equipments and components, are supplied by, and are under the cognizance of, the Bureau of Ships, or such BuShips parts control agency as may be directed.

24-2. RESPONSIBILITY FOR

(1) On board ship the engineer officer is responsible for the care and maintenance of the gyrocompasses and of their spare parts and accessories. (See art. 984, U. S. Navy Regulations.)

(2) On board tenders and repair ships, the repair officer is responsible for the care and maintenance of the gyrocompass repair shop equipment, and for repairs to gyrocompass apparatus.

24-3. SPARE PARTS FOR

(1) Owing to the large number of parts in the gyrocompass, it is impracticable to keep a complete stock of spare parts on board each ship; therefore, a partial supply of spare parts is furnished with each equipment.

(2) The allowance of spare parts on each class of

vessels will be as shown in the allowance list, and as these spares are used, a requisition should be submitted to replace them. Requisitions must show the number on hand, the number allowed, manufacturer's part number, catalog of Navy Material number (when available) and price. These numbers should be carefully verified. The noun description of parts should be that used by the manufacturer or as listed in the catalog. The requisition must indicate whether or not parts required are "In excess" of allowance for the requisitioning activity.

(3) Requisitions for material "in excess" of allowance must be forwarded direct to the Bureau of Ships for approval.

(4) Where spare parts are required to replace surveyed articles, a copy of the survey must be attached to the requisition.

(5) New York Naval Shipyard is the procurement and master control center for repair parts for Gyrocompass and Dead Reckoning, Plotting Facilities and associated equipment. The primary stock of the above repair parts is established at Naval Shipyards New York and Puget Sound. Secondary stocks are established at Naval Shipyard Pearl Harbor and Naval Supply Depot, Guam.

(6) Working stocks of Gyro-compass and Dead Reckoning Equipment repair parts, which are intended primarily to provide for the requirements of the local repair activities and vessels operating in the area, are established at the following locations:

U. S. Naval Shipyards:

Portsmouth, N. H.
Boston
Philadelphia
Norfolk
Charleston

Terminal Island
San Francisco
Mare Island

U. S. Naval Repair Bases:

New Orleans
San Diego

U. S. Naval Operating Bases:

Dutch Harbor
Adak
Subic Bay

U. S. Naval Submarine Bases:

New London
Key West

U. S. Naval Stations:

Guantanamo Bay
Kodiak

(7) Repair activities ashore and afloat located where repair parts are not available from local working stock should obtain parts by requisition to the nearest secondary or primary stocking activity.

24-4. REPAIRS AND ALTERATIONS

All requests for alterations to gyrocompass equipments shall be addressed to the Bureau of Ships. Routine repairs shall be made on approved work requests in the usual manner.

24-5. GYROCOMPASS TEST AND REPAIR FACILITIES

Gyrocompass repair facilities are established at the following activities in addition to those listed in paragraph (6) of article (24-3) above:

U. S. Naval Shipyards:

New York
Puget Sound
Pearl Harbor

U. S. Naval Operating Base:

Guam

U. S. Naval Submarine Base:

Pearl Harbor

All repair ships and tenders are equipped to overhaul and repair gyrocompasses and subsidiary apparatus, including dead reckoning equipments, and carry a limited supply of repair parts. The facilities afloat should be utilized by all vessels to the greatest possible extent.

24-6. GYROCOMPASS SCHOOLS

(1) The class "A" electrical schools at the Naval Training Stations, Hampton Roads, Va., and San Diego, Calif., include in their curriculum an elementary course in the theory, maintenance, and operation of gyrocompasses.

(2) Class "B" gyrocompass schools are maintained at the Naval Training Station, San Diego, and New York Naval Shipyard. Commanding officers are urged to take advantage of these facilities for the instruction of electricians' mates in the theory and maintenance of all types of gyrocompass equipment.

(3) Elementary instruction will be given to electricians' mates in charge of ship's gyrocompass equipment at any gyrocompass repair shop, on application to the officer-in-charge.

(4) When practicable, the ship's gyro electrician's mate shall assist during the overhaul of gyrocompass equipment by any repair forces.

24-7. MASTER GYROCOMPASS OVERHAUL

(1) In the absence of a competent gyro electrician's mate, the master compasses shall not be overhauled except in the presence of an experienced gyro officer, and in no case shall the rotor bearings or balance be changed except by qualified and experienced personnel.

(2) Attention is invited to the fact that gyrocompass equipment is so carefully manufactured and adjusted that repairs should be required very seldom.

(3) The gyrocompass instruction book should be carefully studied. The routine inspections, cleaning, oiling, etc., prescribed in the instruction book, shall be carried out. Repairs shall not be undertaken unless absolutely necessary. An apparent compass error may be due either to an error in computing an azimuth, or to power failure. Before any adjustments to the compass are undertaken, it should be determined that a defect in the mechanism actually exists, and that the unsatisfactory performance is not due to a transient condition.

24-8. INSTRUCTIONS FOR KEEPING SERVICE RECORD BOOK

The instructions for keeping the service record book are outlined in the front of the book and shall be carefully followed. This book is furnished the vessel when the compass is installed, and it should show the history of the compass. The service record book shall always remain with its master compass. The inventory of gyrocompass equipment (Form NavShips 1109) shall be made out from an accurate inventory of the complete equipment.

24-9. GYROCOMPASS LOG AND MACHINERY HISTORY RECORD

In addition to the service record book, there shall be kept a log of the gyrocompass activities and a Machinery History Record of the entire gyrocompass installation. The log shall record the starting and stopping time of each run, and the total hours of operation since delivery. The hours of operation of each electronic tube shall be recorded in a similar manner. Routine cleaning, oiling, and operating conditions and observations shall be entered in the log. In addition to the entries required in the service record book, all repairs, alterations, adjustments, and tests of the entire gyrocompass installation shall be entered in the Machinery History Record.

24-10. INSPECTIONS

(1) Gyrocompass officers attached to tenders, repair ships, and the various staffs should visit, at every opportunity, ships of their respective forces for the purpose of inspection of equipment, and for the instruction of personnel in the care and operation of compasses. Gyrocompass officers should impress on

personnel the importance of keeping their spare-parts boxes complete; and also that, due to the excessive cost of spare parts, great care should be exercised against using the parts unnecessarily.

(2) A report of any unfavorable conditions found shall be made to the commanding officer of the vessel concerned for his information.

(3) No report of inspection need be made to the Bureau except in such cases as may furnish information in regard to design, operation, or maintenance which it is thought would be of interest or value to the Bureau or the service at large.

(4) Every gyrocompass installation, including all subsidiary equipment, apparatus, appurtenances, and spare parts, shall be inspected once each quarter by a competent gyrocompass officer.

24-11. REPORTS (NAVSHIPS 1109)

(1) All vessels shall submit Form NavShips 1109 annually on 31 May.

(2) These forms shall be submitted upon installation or removal of gyrocompass or auxiliary equipment.

Part 2—Magnetic Compasses

24-21. DESCRIPTION OF NAVY MAGNETIC COMPASS AND BINNACLES

Descriptions of the various Direct and Remote Reading Magnetic Compass and Binnacles for Naval Vessels are delineated in the Appendix to CNO directive Op23(QB/IC)S24-7 Serial No. 911223 dated 2 August 1944 on Standard Magnetic Compass Installations.

24-22. ALLOWANCES

The number and types of Magnetic Compasses allowed for all types of Naval vessels are listed in enclosure (A) to CNO directive Op23(QB/IC)S24-7 Serial No. 911223 dated 2 August 1944 on Standard Magnetic Compass Installations.

24-23. SUPPLY

(1) Direct Reading Magnetic Compasses are stocked at activities listed in paragraph 24-31(1) and 24-43(1) of this Manual.

(2) Remote Reading Compasses are stocked at Material Laboratory, New York Naval Shipyard, Brooklyn, N. Y., but are issued only by direct authority of the Bureau of Ships.

(3) Descriptive information pertaining to all Magnetic Compasses except remote reading is contained in Class 18 Material Section of the Catalog of Navy Material—General Stores Section.

24-24. INSTALLATION

Magnetic Compass installations in Naval vessels are made under the direction of the Bureau of Ships. In order to obtain satisfactory Magnetic Compass performance it is necessary that the immediate vicinity of the Direct Reading compasses and transmitter

of the Remote Reading compasses be free of any material, fixed or movable, which will introduce permanent or transient magnetic fields at the element. Accordingly magnetic materials in the vicinity of the compass locations are limited as specified by Section C-6-23 and Subsection C-6-g of Section C6-22 of General Specifications for Building Vessels of the U. S. Navy. Ships' officers should assist the Bureau of Ships by reporting existing and continuing violations of these specifications.

24-25. REPAIR OR REPLACEMENT

(1) Minor repairs to Direct Reading Magnetic Compasses may be accomplished at the Instrument Shop at any U. S. Naval Shipyard or aboard Naval Repair Ships. Major repairs to Direct Reading Magnetic Compasses are made at the U. S. Naval Observatory. Magnetic Compasses requiring major repairs or those found defective should be replaced.

(2) Repairs to all Remote Reading Magnetic Compasses are made at the Material Laboratory, New York Naval Shipyard, Brooklyn, New York. Minor repairs may be effected under the jurisdiction of a Compass Officer. Should it be deemed necessary to remove a Remote Reading Magnetic Compass for repair or replacement, all main components should be removed and shipped as a unit.

(3) Wherever and whenever possible Magnetic Compass personnel located at any of the several major Naval Shipyards or bases should be consulted in matters pertaining to the repair or replacement of Direct or Remote Reading Magnetic Compasses.

24-26. MAINTENANCE OF MAGNETIC COMPASSES

Chapters 22 and 26 of U. S. Navy Regulations delineate responsibilities for maintenance of Magnetic Compasses.

24-27. ADJUSTMENT AND COMPENSATION

Magnetic Compass Officers are located at all major U. S. Naval Shipyards and most U. S. Naval Bases to adjust and compensate, or assist in the adjustment and compensation, of magnetic compasses for both normal and degaussed conditions of Naval Vessels. However, adjustment and compensation instructions may be found in any of the following publications:

"Handbook of Magnetic Compass Adjustment and Compensation"—Hydrographic Office Publication No. 226 by Spencer and Kucera.

"Navigation and Nautical Astronomy" by Dutton.
"A Treatise on Compass Compensation" by Kielhorn.

Special assistance on, or special training in, the more unusual problems of Magnetic Compass Adjustment and compensation may be obtained upon request to the Bureau of Ships, Navigational Equipment Section.

24-28. OBSERVATIONS AND RECORDS

Magnetic Compass observations are to be made and reported on at such times as outlined in Chapters 26

and 54 of U. S. Navy Regulations. Detailed instructions relative to compass observations and the maintenance of records also appear on the inside of the covers of the Magnetic Compass Record Book (NAV SHIPS 1101) which is maintained aboard every U. S. Naval Vessel in commission.

24-29. REPORTS

Reports on Magnetic Compasses shall be made as required in Chapter 54 of U. S. Navy Regulations.

Part 3—Chronometers and Watches

24-31. SUPPLY AND REPLACEMENT

(1) The Naval Observatory issues ship's chronometers, gimbale navigatng watches, and nongimbale navigatng watches. Except as replacement, these instruments are not subject to requisition or survey, but are issued only upon the Bureau of Ships approval of a request by letter. When a chronometer, gimbale navigatng watch, nongimbale navigatng watch, navigator's comparing watch, or navigator's stop timer stops or develops an erratic rate, it shall, without reference to the Bureau, be exchanged as soon as practicable for a similar instrument at the nearest authorized distributing pool for such instruments. Chronometer pools are located at the following activities:

Naval Shipyards:

Mare Island
Boston
Charleston
New York
Puget Sound
Philadelphia

Naval Supply Depot:

Norfolk
Balboa, C. Z.
Pearl Harbor
San Diego
San Pedro

Naval Station:

Key West

Naval Supply Pier:

Miami, Florida (Chronometers and watches only)

Naval Observatory:

Washington, D. C. (Main Supply Point)

These pools carry a stock of ship's chronometers, gimbale navigatng watches, nongimbale navigatng watches, comparing watches, and stop timers.

(2) Comparing watches and stop watches are subject to requisition and survey.

(3) In case of loss, the procedure stated in article 24-42 should be followed.

24-32. OVERHAUL

Chronometers shall be turned in to the Naval Observatory or the nearest authorized distributing pool for replacement within 3 years from the date of last

overhaul. Gimbale navigatng watches and nongimbale navigatng watches shall be turned in to the Naval Observatory or the nearest authorized distributing pool within 2 years from the date of last overhaul. Except in emergencies, none of these instruments shall be maintained in service more than the authorized period of time. Replacements shall be accomplished without reference to the Bureau. The Naval Observatory is the sole activity authorized to effect the overhaul, repair, regulation, and compensation of all Bureau of Ships chronometers, gimbale navigatng watches, and nongimbale navigatng watches. The overhaul, repair, regulation, and compensation of Bureau of Ships chronometers, gimbale navigatng watches, and nongimbale navigatng watches shall not be undertaken by any other activity or by commercial activities. Vessels whose records do not show the last date of overhaul of chronometers, gimbale navigatng watches, and nongimbale navigatng watches may obtain these dates from the Naval Observatory.

24-33. CARE AND HANDLING

Complete instructions relative to the care and handling of chronometers, gimbale navigatng watches, and nongimbale navigatng watches are contained in the Ships Chronometer Record Book (Form NavShips 702).

24-34. SUBMERGED TIMEPIECES

If a chronometer, a gimbale navigatng watch, or a nongimbale navigatng watch has been submerged in water, it shall be completely opened, drained, and immersed in glycerine for at least 1 hour. It shall then be tightly closed with as much glycerine as possible retained in the case, carefully packed, and shipped at the earliest practicable time to the Naval Observatory for prompt and proper action. Comparing watches and stop timers shall be handled in a similar manner but they may be reconditioned by any authorized watch repair activity.

Part 4—Care, Custody and Replacement of Navigational Instruments

24-41. CARE OF NAVIGATIONAL INSTRUMENTS

(1) Great care is necessary to prevent damage to navigational instruments, especially sextants, stadimeters, binoculars, spyglasses, and ships' telescopes. Examples of injury to these instruments are: Sextants and stadimeters misaligned due to rough usage and careless handling or dropping; ships' telescopes knocked over due to pedestal not being secured to the deck; spyglasses and binoculars being left where they become wet by rain or salt water, and being put away while still damp; spyglasses and binoculars being placed where the roll of the ship causes them to fall on the deck; lenses and prisms scratched and parts lost or improperly assembled due to efforts by unskilled persons to clean or repair instruments.

24-42. CUSTODY AND LOSS OF WATCHES AND BINOCULARS

(1) Stop watches, comparing watches, and binoculars shall be issued only to an officer or an authorized petty officer. Custody receipts shall be required in all cases. The individual signing the custody receipt will be held responsible for the loss of any of these articles unless he can show that he had taken reasonable precaution to safeguard the property and that the loss was not due to his carelessness. A simple statement that the article is missing or was stolen by some unknown person will not relieve the custodian of the responsibility for the loss of the article.

(2) When the loss of any of these articles becomes known, the Commanding Officer shall institute a rigorous search on board ship. If the article is not found, he shall, where theft may be suspected, communicate with the local police authorities, giving serial numbers, make, mark and modification of the lost article, with a view to its recovery in case it is offered for sale or pawned.

24-43. SUPPLY AND REPLACEMENTS

(1) In addition to chronometers and navigating watches listed in paragraph 24-31 (1) the Bureau of Ships is responsible for the supply and replacement of the following instruments:

- *Binoculars
- *Compasses
- *Sextants
- *Watches, comparing
- *Watches, stop timers (Navigational only)
- C. I. C. equipment

*Listed in the Catalog of Navy Materials—General Stores Section under complete control of the Bureau of Ships.

(2) The supply and replacement of all other navigational equipment is the responsibility of the Bureau of Supplies and Accounts.

(3) For replacement of damaged navigational equipment and for issues not "in excess" of navigational instruments listed in paragraph 24-43(1) moderate stocks are maintained at all activities listed in paragraph 24-31(1) and the following:

Naval Shipyards:

Norfolk
Portsmouth

Naval Supply Depots:

Bayonne
Guantanamo Bay, Cuba
Oakland
Scotia, New York
Seattle

(4) In addition to the above, navigational instrument emergency pools are maintained aboard ARH, AR, AD and AS vessels and at advanced optical repair shops. These instruments are for use for exchange purposes when a vessel's availability is too short to allow adequate repairs. Instruments shall be exchanged and instruments needing repairs shall be repaired at the convenience of the repair activity and returned to the emergency pool. When an in-

strument has to be surveyed, a replacement may be obtained from the emergency pool. The vessel receiving the replacement shall then originate a "not in excess" requisition for an instrument (with reference to this paragraph) to be delivered to the repair activity emergency pool.

	ARH	AR	AD	AS	Base
Alidade, Surface Lookout.....	X	X	X		X
Alidade, Telescopic.....	X	X	X	X	X
Azimuth Circle.....	X	X	X	X	X
Azimuth Telescope.....	X	X			
Barometer, Aneroid.....	X	X	X	X	X
Barometer, Mercurial.....	X	X			
Binoculars, 7x50.....	X	X	X	X	X
Chronometer, Standard.....	X	X	X	X	X
Circle, Bearing.....	X	X	X	X	X
Clock, Deck.....	X	X	X	X	X
Clock, Mech. "A", 6".....	X	X	X	X	X
Clock, Mech. "B", 6".....	X	X	X	X	X
Clock, Mech. "A", 8½".....	X	X	X		X
Clock, Mech. "B", 8½".....	X	X	X		X
Compass, Armored Vehicle w/spheres & brackets.....				X	
Compass, Boat, w/binnacle.....	X	X	X		X
Compass #1.....	X	X	X		X
Compass #2.....	X	X	X		X
Compass #3.....	X	X			
Indicator, Position Angle.....	X	X	X	X	X
Instruments, Drawing w/acc....				X	
Manometer, Aneroid.....				X	
Protractor, Parallel Motion.....	X	X	X	X	X
Sextant, E.T.S.....	X	X	X		X
Spyglass, O.O.D.....	X	X	X		X
Spyglass, Q.M.....	X	X	X		X
Stadimeter.....	X	X	X	X	X
Telescope, Ships'.....	X	X			
Watch Comparing.....	X	X	X	X	X
Watch, Gimballed Navigating...	X	X	X	X	X
Watch, Nongimballed Navigating	X	X	X	X	X
Watch, Stop Timer.....	X	X	X	X	X

(5) Defective instruments shall be turned in to the nearest repair activity. Replacements shall be obtained from the emergency pool of the repair activity or the nearest supply activity (see paragraph 24-43(3)). When an instrument is turned in for replacement, it shall be expended from the Ship's Equipage Records and taken up by the receiving activity in Appropriation Purchases Account as a receipt from the Ship's Title "B", and from the accounting number of the vessel. Title "B" navigation instruments turned in for replacement by a ship shall be accompanied by priced invoices.

24-44. REPAIRS

(1) Except as noted further in this paragraph, repairs to navigational instruments shall be performed by: (1) the Naval Observatory, (2) Naval Shipyards, (3) Advanced Base Shops, (4) Repair Base Shops, and (5) Repair Ships and Tenders.

(a) Mercurial barometers shall be repaired only at

the Naval Observatory and Naval Shipyards: Mare Island, Puget Sound and Pearl Harbor.

(b) Chronometers, gimballed navigating watches, and nongimballed navigating watches shall be repaired and regulated only at the Naval Observatory.

(c) Major repairs to sextants (those repairs involving replacement or realignment of the arc, the arm, the center, the graduated scale, or the worm sector insert are considered major repairs) shall be done only by the Naval Observatory. If any doubt exists as to the accuracy of the sextant, it shall be returned to the Naval Observatory for collimation.

(2) If an instrument is determined, by the repair activity, to be obviously beyond economical repair, it should be expended in accordance with current instructions for survey of Title "B" equipment, and a replacement instrument requisitioned as a "not in excess" item according to the vessel's allowance. Surveyed instruments shall be turned over to a repair activity for disposition. The repair activity will dismantle the instrument for salvage of usable parts or dispose of the instrument locally according to the decision of the repair officer and/or optical officer.

(3) Navigational equipment repair parts for those instruments for which repair units are carried in stock shall be requisitioned from the Bureau of Ships or from distributing activities as designated by the Bureau of Ships. Repair parts will be issued only to authorized repair activities.

(4) Navigational instruments from the emergency pool may be used for exchange purposes, at the discretion of the commanding officer, the repair officer, and the optical officer, when a vessel has a short

availability or it becomes necessary to distribute the work load between availability of vessels.

24-45. ALLOWANCE LISTS AND INVENTORIES

(1) Lists of Navigational Instruments, Title "B", equipage, and title "C", supplies, are issued on form NavShips 1428 or in Group S24-7, Part II of the Bureau of Ships Allowance Lists which are prepared in accordance with Group L7 of the Bureau of Ships Master Allowances. This allowance list is issued by the Bureau for all vessels of the Navy and, in addition to the navigational instruments, includes such materials as surface lookout equipment; visual aids for lookouts, gunnery personnel, and signal force; mechanical clocks; C.I.C. equipment, etc.

(2) When additional navigational equipment is considered necessary because of the vessel's construction or special duty, "in excess" requisitions shall be submitted to the Bureau stating purpose and reason for which the additional instrument is required and no change of allowance requested. Changes of allowance shall be applicable to all vessels of the same class; requests from them shall, therefore, be forwarded via the type commander.

(3) All vessels shall submit form NavShips 1108 (Inventory of Navigational Instruments) upon request of the Bureau.

24-46. CHEMICAL SOUNDING TUBES

Chemical sounding tubes which are not hermetically sealed at both ends or which have been aboard ships longer than three years shall be destroyed and replaced.

Section II. Periscopes

Part 1—General

24-51.

Periscopes are considered to be a part of the ship's outfit and are carried on the allowance lists as equipage under title "B."

24-52. NOMENCLATURE

To insure a uniform method of designating periscopes on submarines, the following system of nomenclature should be used in connection with all correspondence, specifications, and plans relating to such instruments.

24-53.

The periscope nearest the bow will be called No. 1 periscope, irrespective of its type or the location in the vessel of the eyepiece. The next periscope aft of No. 1 periscope will be called No. 2 periscope, and the next periscope aft of No. 2 will be called No. 3 periscope. The terms "forward," "middle," and "after" periscopes or "first," "second," and "third" periscopes will not be used in designating these instruments.

24-54. DEFINITIONS

(1) *Periscope*.—The term "periscope" is used generally to designate all types of instruments, but specifically to designate instruments which are designed for horizontal view only.

(2) *Altiscope*.—An instrument for zenith view only.

(3) *Altiperiscope*.—Instruments having the combined qualities of altiscopes and periscopes, sometimes called altiscope-periscopes and sometimes altiazimuth instruments, shall be called altiperiscopes. All periscopes now in service in submarines are altiperiscopes.

(4) *Unifocal and bifocal*.—Instruments of single and double powers, respectively.

(5) *Night periscope*.—A periscope having a high light transmission in conjunction with a large diameter exit pupil.

(6) *Attack periscope*.—A periscope designed with a minimum diameter of head at the sacrifice of light transmission and diameter of exit pupil.

24-55. REGISTRY NUMBERS

Periscopes are numbered serially by the Bureau in order of their manufacture, such numbers being referred to as "Bureau of Ships Registry No." It is essential to mention the registry number in all correspondence referring to particular periscopes.

24-56. DESIGN DESIGNATIONS

(1) Each separate or modified design of periscope will be assigned a design designation which should always be reported in all correspondence concerning periscopes, in addition to the registry number. This design designation will be assigned by the Bureau and will consist of the following parts in the order given:

(a) A serial number for each design will be assigned by the Bureau.

(b) A letter to indicate the manufacturer:

K for Kollmorgen.

E for Keuffel & Esser.

B for Bausch & Lomb.

(c) A letter or letters to indicate the type of periscope:

A. Bifocal altiperiscope.

N. Night or low-visibility altiperiscope.

(d) A number equal to the optical length of the instrument in feet to the nearest foot.

(e) For a period, the letter T was added to indicate that the optics of the instrument had been treated to increase light transmission and improve definition. Since all periscopes in service have been so treated and new periscopes are furnished treated, this letter is not being included in the recent design designations.

(f) If the outer diameter of the upper portion of the reduced head section is less than 2 inches, a number representing the outer diameter of the upper part of the reduced head section in inches will be added, separated from the preceding character by a diagonal mark.

(g) If the instrument is an altiperiscope designed to permit view at any angle from the zenith to a point below the horizon, the letters HA will be added.

(2) As an example the following is quoted:

89 (serial number).

K (Kollmorgen).

A (bifocal altiperiscope).

40 (optical length in feet to nearest foot).

T (treated optics).

1.414 (outside diameter of upper part of reduced head section in inches).

HA (high angle).

Combined, this design designation reads as follows:

89KA40T/1.414HA

Part 2—Installation**24-61. CARE IN INSTALLING**

Particular care must be exercised in the initial installation of periscopes. The alinement of bearings

must be done with the greatest degree of accuracy. The quick and efficient use of the instrument depends greatly upon the freedom of training the instrument in azimuth. The clearance between the periscope tube and the bearings must be minimized in order to reduce vibration. It should be between 0.003 and 0.007 inches.

24-62. EXAMINATION AND TESTS

If at all practicable, when new periscopes are installed, submerged tests should be made as soon as possible after the installation is completed in order that the airtight and watertight integrity of the installation may be tested under actual conditions.

Part 3—Handling and Storage**24-71. CARE IN HANDLING**

Great care must be exercised in the handling and installation of periscopes. A part of the trouble experienced with these instruments can be traced directly to carelessness. Rough and inefficient handling may easily bring about the following:

(1) Bent or distorted tube.

(2) Leaky joints.

(3) Chipped and broken lenses.

24-72. HANDLING DURING INSTALLATION

The periscope shall be hoisted for installation by means of a clamp of ample proportions with an accurate bore. A bronze or bronze-lined clamp is preferable, although a steel clamp may be used. The clamp shall be hung from two chains of sufficient length and provided with a spreader to prevent contact with the relatively fragile head and reduced section. The clamp shall be placed around the body tube at least 12 inches below the joint between the tube and the reduced section. One or two safety clamps (the brass clamps supplied with the shipping box will serve) shall be secured above the hoisting clamps. Under no circumstances shall clamps containing set-screws be used, nor should any clamp be placed directly over the joint between the body tube and the reduced section, as otherwise severe damage to the periscope may result.

24-73. CARE WHILE IN STORE

Where periscopes lie in store, the instruments, or their containing boxes, must be properly supported, with no overhanging ends. In unpacking and handling the instruments, they shall be supported at several points so that no bending stress is placed on the instruments or the joints thereof. Instructions for routine care and inspection of periscopes in store are included under chapter 9 of this manual.

24-74. INSTRUCTIONS TO MECHANICS

Instructions shall be issued to all employees and enlisted personnel performing work on periscopes, or on apparatus in the immediate vicinity of periscopes, that extreme care must be used in carrying out their

work. Supervising mechanics and petty officers shall be especially cautioned and made responsible that mechanics under their supervision comply with instructions.

24-75. SHIPPING BOXES

Shipment of periscopes must in all cases be made in boxes specially constructed to provide adequate support for the instrument and to insure it against severe shocks. Wherever possible, shipments should be made in the boxes originally furnished by the contractor. Shipments must be marked to indicate that they are delicate instruments and require careful handling.

24-76. SHIPPING BOX SERIAL NUMBER

The Bureau will assign each periscope shipping box or container a serial number which shall be inscribed on the shipping box label plate.

24-77. SERIAL NUMBER TO BE REPORTED TO THE BUREAU

The Bureau of Ships serial number of the periscope shipping boxes involved shall be reported to the Bureau in each of the following cases:

(1) When periscopes are shipped from one point to another, the Bureau of Ships registry number of each periscope and the corresponding Bureau of Ships serial number of its shipping box shall be stated on the bill of lading and the endorsement on the salmon copy of the shipping order reporting that shipment has been made.

(2) When periscopes are installed on submarine at building yards and naval stations, or by a base or tender force, the Bureau of Ships serial number of the shipping box from which each periscope was removed shall be stated in the report of periscope installation.

(3) When empty boxes are turned into store from any ship or building yard.

(4) When empty boxes are shipped from one point to another.

(5) When shipping boxes are no longer serviceable, they shall be scrapped and a report giving the serial number shall be forwarded to the Bureau.

24-78.

In the case of boxes on which no Bureau of Ships serial number appears or for which no number has as yet been assigned, an identifying symbol shall be inscribed on the box prior to its shipment or other disposition and the Bureau shall be informed about the over-all dimensions of the box and the identification mark.

Part 4—Reports

24-91. REPORTS

When a submarine is commissioned, and when a new periscope is installed, a special report, contain-

ing the following data, shall be submitted to the Bureau by the commanding officer:

(1) Registry number, design designation, and location of each periscope installed in the vessel.

(2) Any defects and any outstanding features in the design of the periscope which should, in the opinion of the commanding officer, be altered in that design of periscope or in subsequent designs of periscopes.

(3) Any mechanical or optical defects not apparently inherent in the design which exist in any periscope installed in the vessel. In this connection comment should be submitted in regard to hard operation of the power shift, focusing gear, prism tilt, or other moving parts.

(4) Any additional comment which may appear pertinent.

24-92.

Whenever possible, information of special interest regarding the relative efficiency, either optical or mechanical, of different periscopes installed in a submarine shall be submitted to the Bureau for its information in the design of new periscopes.

24-93.

When practicable, reports (art. 24-91) shall be submitted in the form of force, squadron, or division reports rather than as separate reports from individual vessels.

24-94. SEMIANNUAL REPORTS

Attention is invited to the next to the last paragraph of form NavShips 179 which is used for semi-annual reports of hull and hull material for submarines. The information required by this paragraph shall be submitted in full on all semiannual hull inspection reports for submarines. The information in regard to the registry numbers of the periscopes installed in the vessel and the location of the periscopes by registry numbers is of especial importance. In addition to the registry number, the design designation must, in the case of each periscope to which a design designation has been assigned, be included.

Part 5—Care and Maintenance

24-101. BENDING, VIBRATION, AND SHOCK

(1) The necessity for protecting submarine periscopes from damage due to the bending stress and vibration when the vessel is running submerged and to the vibration caused by the engines when the vessel is running on the surface cannot be too greatly emphasized. When operating submerged, periscopes should not be extended to a distance which will cause excessive bending or vibration at the speed at which the vessel is running. The speed at which the vessel can operate with any periscope fully extended varies in different cases, and this speed should be determined by the commanding officer separately for

each individual periscope. In general, conditions which tend to cause any appreciable bending or vibration of a periscope should, as far as practicable, be avoided. When the vessel is proceeding on the surface, the periscope should be habitually kept in the position in which engine vibration is minimized.

(2) In order to minimize the damage which may result from the shock of underwater explosions produced by depth charges, the periscope should be fully housed during such attacks.

24-102. PERISCOPE USED FOR SPOTTING

While the Bureau does not desire to prohibit the use of periscopes for spotting during gun firing, commanding officers must realize that under such conditions they are subjecting the instruments to stresses for which they were not designed and that defects are liable to develop if the practice is continued for extended periods.

24-103. TURNING IN AZIMUTH

To insure proper lubrication and satisfactory operation of the hoist, periscopes shall be raised daily to their full height, rotated through the full arc of the train and lowered.

24-104. RESPONSIBILITY FOR DAMAGE

If a periscope is damaged because the precautions of articles 24-101 to 24-103, inclusive, have not been followed, then the responsibility for such damage rests with the officer or other person responsible for exercising these precautions.

24-105. FOGGING

Fogging may be either external or internal. (1) External fogging of the lower eyepiece window occurs at times when the periscope is first raised after it has been housed in a cold periscope well for a long period. This fogging can be wiped off with lens paper and will cease forming when the instrument warms up. The only other instance of external fogging is that of the outside of the top staunching window which can occur when the temperature of the sea water is appreciably colder than the dew point of the outside air. This type of fogging is recognized by the fact that it is not apparent when the periscope is first raised but slowly appears as the surface of the window dries. It can be eliminated by occasionally lowering the periscope and wetting the window.

(2) Internal fogging occurs on one or more optical surfaces whenever the temperature of the optics falls below the dew point of the gas with which the periscope is charged. It is prevented by charging the periscope with nitrogen of which the dew point is less than the lowest temperature met by the periscope in service.

24-106. DRYING AND RECHARGING

Periscopes should be dried out and recharged as outlined in Sections 24-107 and 24-108 under the following circumstances:

(1) When the internal pressure has fallen to 4 p.s.i. or less or is released for any reason.

(2) When the dew point is found to be above minus 50°C.

(3) When fogging is reported, if internal fogging is indicated.

24-107. PREFERRED METHOD OF DRYING AND RECHARGING

The preferred method of drying and recharging, which should, where possible, be undertaken only by personnel trained in the process, is described below. It should only be performed where the temperature of the periscope is above 50°F.

(1) The equipment required consists of the following:

A high vacuum gauge of the McLeod type or a suitable manometer.

A high vacuum pump, "Cenco Hyvac" or equal.

A supply of solid carbon dioxide.

A supply of dry nitrogen.

A supply of acetone, technical grade.

A dew point tester, as described,

together with copper tubing, rubber vacuum hose, and other apparatus commonly available.

(2) The periscope should first be subjected to an internal pressure test, introducing nitrogen slowly over a two-hour period through the air inlet valve until a pressure of 100 pounds is reached. The instrument should then be soaped or preferably immersed in water, and minutely examined for leaks, particularly through packing glands. All leaks must be eliminated, otherwise the high vacuum cannot be obtained. The use of wax to stop leaks is not acceptable except as a seal over the stems of the air valves.

(3) The pressure is then slowly released over another two-hour period, and the air outlet valve of the periscope connected to the vacuum pump through the minimum practicable length of hose. The vacuum gauge or manometer is similarly connected to the air inlet valve. The pump should be operated until a vacuum of preferably 2 m/m of mercury (and in no case more than 4 m/m) is attained. From three to six hours is usually required to obtain the desired vacuum. If only an insufficient vacuum is obtained, which fails to improve with continued pumping, it may be assumed that leaks are present.

(4) When the required vacuum is reached, the vacuum line should be closed and the periscope allowed to stand for three hours or more. If a slight rise in pressure occurs, it will be caused by residual moisture, and further pumping will be required, following which the vacuum line may be removed and replaced with the nitrogen supply line for final charging. Nitrogen should not be taken from a flask in which the pressure is less than 400 p.s.i.

(5) A dryer must be placed in the line between the nitrogen flask and the periscope. The preferred form of dryer comprises a coil of copper tubing immersed in a mixture of solid carbon dioxide and acetone

which is at a temperature of approximately minus 78°C. A filter is desirable to trap entrained water and dust particles. The dried nitrogen should be fed in slowly until the periscope attains an internal pressure of 10 pounds.

(6) The dew-point tester is a small Erlenmeyer flask with the bottom externally silvered. The tester should be connected to the air outlet valve, and should be partially immersed in a beaker of acetone to which solid carbon dioxide is slowly added to reduce the temperature progressively. The gas from the periscope should be led into the flask to a point within $\frac{1}{2}$ inch from the silvered surface of the bottom. The temperature of the acetone should be carefully noted at the moment the first trace of condensation occurs upon the silvered surface. A dew point at or below minus 50°C. is satisfactory; in case a higher dew point is noted the vacuum should be repeated and the periscope recharged. Following the dew point test the gas in the periscope in excess of a pressure of $7\frac{1}{2}$ pounds should be bled off, and the instrument sealed.

(7) In the case of a periscope in which the pressure has fallen below 7 p.s.i., but is over 4 p.s.i., recharging only may be necessary. The dew point should be checked to determine that no moisture has entered, and the instrument should be examined for hermetical tightness. Nitrogen dried in the manner previously described should be introduced until a pressure of 10 pounds is reached. Another dew point test should be made and the pressure reduced to $7\frac{1}{2}$ pounds.

(8) As the internal pressure affects the optical properties of the instrument, pressures in excess of $7\frac{1}{2}$ p.s.i. should not be used unless necessary to obtain a satisfactory focus.

(9) All apparatus used in drying a periscope should be kept clean and employed for no other purpose. The dew point tester should be dried out before each use.

24-108. USE OF SILICA GEL DRYER

At bases and tenders where no solid carbon dioxide is available, the procedure outlined in article 24-107 is modified to use a silica gel dryer (Bureau of Ordnance, Mark 3, Mod O Instrument Dryer with a filter to prevent particles of the gel from entering the periscope) for drying the nitrogen. For each charging operation the silica gel must be reactivated by heat-

ing in a covered kettle or pan to 480°-500°F. for two hours. As the lack of solid carbon dioxide precludes the use of the dew point tester, special care must be employed to exclude moisture.

24-109. TAGGING AFTER DRYING

After a periscope has been dried and recharged by one of the methods specified in article 24-107 or 24-108, a tag, stating the date of drying and the pressure to which the instrument was charged, should be attached to the periscope.

24-110. EMERGENCY DRYING

In an emergency it may be possible to dry out the periscope if a supply of dry nitrogen or nitrogen-air mixture is available. The gas supply should be connected to the inlet valve, the plug in the outlet valve removed, and the gas permitted to circulate gently for 30 minutes (this depends on amount of dry gas available). The outlet valve should then be closed and the pressure permitted to build up to $7\frac{1}{2}$ pounds. This procedure should only be used as an expedient until it is possible to dry out the periscope by one of the methods outlined in articles 24-107 and 24-108.

24-111. THE USE OF CALCIUM DUST OR OTHER FOREIGN MATTER IN THE PERISCOPE AS A DRYING AGENT IS PROHIBITED

24-112. INTERNAL PRESSURE

When a periscope is installed, the installing activity should check the pressure in the instrument. If the pressure is below 7 p.s.i., the instructions given in article 24-107 should be followed. After installation the pressure should be tested at least once a month.

24-113. CARE OF FLOODED PERISCOPES

When periscopes have been flooded and subsequently emptied of water, the resulting corrosion will damage certain parts beyond repair unless prompt countermeasures are taken. In the event of flooding the procedure described below should be followed:

(1) As soon as possible after flooding, flush with fresh water and dry thoroughly, if practicable. Then apply a rust preventive compound similar to Tectyl Grade III or No. 511 to the internal parts.

(2) If Tectyl is not available, the periscope should be sealed after flushing and left full of fresh water until repairs are effected.

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